Day 24

UST – Employee Skills & Project Tracking System (ESPTS)

Project Requirement Document (MongoDB-based)

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1. Project Overview

UST wants to build an internal **Employee Skills & Project Tracking System (ESPTS)** to maintain accurate information about employees, skills, departments, project allocations, and experience levels.

This system will be used by:

- HR team
- Delivery Managers
- Resource Allocation team
- Internal L&D team

The backend database for this system must be built using **MongoDB**, utilizing its flexible document model.

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2. Purpose of the System

The goal is to create a **single MongoDB collection** that stores **employees and their related operational data**, allowing for:

- Easy insertion of new employee profiles
- Searching / filtering based on skillsets, department, experience
- Updating employee competencies and project status
- Archival and deletion operations for outdated records

This project MUST demonstrate full coverage of **CRUD operations** (Create, Read, Update, Delete) in MongoDB.

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3. Target Collection

The system will maintain a single MongoDB collection:

```
employees
```

Each document represents a UST employee.

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4. Functional Requirements

The system MUST support the following MongoDB operations:

4.1 Create Operations

- Insert new employee profiles individually
- Bulk insertion of multiple employees
- Create nested fields
- Store project details as an array of objects
- Store skills as an array of strings

4.2 Read Operations

- Find employees by department
- Find employees by skillset
- Find employees by city
- Read employees missing particular fields (e.g., skills missing)
- Filter by experience range
- Query nested fields (e.g., project-status)
- Projection for limited fields

- Sorting by age, experience, name

4.3 Update Operations

- Modify employee department
- Add or remove skills
- Update nested project status
- Add new project entries
- Increment experience counter
- Rename and delete fields
- Bulk update based on conditions
- Upsert employee profile if not found

4.4 Delete Operations

- Delete individual employee profiles
- Delete employees matching conditions (e.g., low experience)
- Delete specific fields (not entire document)
- Remove array elements (skills, projects)

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5. Employee Document Structure

Every employee record MUST follow below structure (fields may vary across employees):

```
{
  "emp_id": <Number>,
  "name": <String>,
  "age": <Number>,
  "department": <String>,      // Examples: "AI", "Cloud", "Testing", "Cybersec
urity"

  "skills": [<String>],        // List of competencies
```

```
  "address": {                // Nested document
    "city": <String>,
    "state": <String>
  },

  "projects": [               // Array of objects
    {
      "name": <String>,
      "status": <String>       // "planned", "in-progress", "completed", "deploye
d"
    }
  ],

  "experience_years": <Number>  // 0–30
}
```

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6. Detailed Requirements

6.1 CREATE Requirements

6.1.1 The system must allow insertion of new employees

- Records can be inserted one-by-one and in bulk
- Some employees intentionally have missing fields (e.g., no skills, no projects)

6.1.2 Each inserted employee must be unique by `emp_id`

6.1.3 The database must automatically create the collection on first insert

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6.2 READ Requirements

6.2.1 The system must support querying by:

- Department
- Skill
- City / State
- Age range
- Experience level
- Employees missing “skills” field
- Employees with specific project status
- Employees working on >1 project

6.2.2 Results must support:

- Ascending/descending sorting
- Field projection (retrieve only selected fields)
- Filtering nested documents
- Handling arrays (finding by element value)

6.3 UPDATE Requirements

The system must support ALL major MongoDB update operations:

6.3.1 Update fields using `$set`

(E.g., change department of an employee)

6.3.2 Increment values using `$inc`

(E.g., increase experience for all employees annually)

6.3.3 Remove fields using `$unset`

(E.g., remove address.state for all employees)

6.3.4 Update arrays:

- Add new skills (`$push`, `$addToSet`)

- Remove skills (`$pull`)
- Update project statuses using positional operator (`$`)

6.3.5 Update nested fields

(E.g., modify project status inside `projects` array)

6.3.6 Bulk update using `update_many`

(E.g., mark all cloud employees as "Cloud-Engineer")

6.3.7 Upsert unregistered employees

- If `emp_id` does not exist, create a new entry automatically

6.4 DELETE Requirements

System must support:

6.4.1 Delete employee record

- Precise deletion using `emp_id` (delete_one)
- Department-based deletion (delete_many)

6.4.2 Delete fields within employee doc

(e.g., remove department field from all employees — simulated cleanup)

6.4.3 Remove array entries

(E.g., remove a project or specific skill)

6.4.4 Delete employees with:

- No skills
- Experience < 1 year
- No ongoing projects

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7. Non-Functional Requirements

7.1 Data Consistency

Employees must be uniquely identifiable via `emp_id`.

7.2 Flexibility

Documents may vary in structure (MongoDB dynamic schema).

7.3 Maintainability

Every update and delete operation must be reversible using backups.

7.4 Simplicity

The project must use **basic MongoDB**, not advanced drivers or ORMs.

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8. Expected Deliverables

1. **Database Setup**
    - One collection: `employees`
    - 25 sample documents meeting required structure
2. **CRUD Implementation**
    - Full PyMongo script executing all mandatory CRUD operations
3. **Project Report (optional)**
    - Summary of operations performed
    - Screenshots of Compass views
    - Python code snippets

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9. Completion Criteria

The project is considered complete when the trainee can:

✔ Insert flexible employee records

✔ Retrieve documents in complex ways

✔ Update nested structures, arrays, and multiple documents

✔ Use all essential MongoDB update operators

✔ Delete documents and their fields safely

✔ Demonstrate confidence in CRUD operations

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10. JSON Sample Dataset (10 Employees)

```
[
  {
    "emp_id": 101,
    "name": "Anu Joseph",
    "age": 23,
    "department": "AI",
    "skills": ["python", "mongodb"],
    "address": { "city": "Trivandrum", "state": "Kerala" },
    "projects": [
      { "name": "VisionAI", "status": "completed" },
      { "name": "DocScan", "status": "in-progress" }
    ],
    "experience_years": 1
  },
  {
    "emp_id": 102,
    "name": "Rahul Menon",
    "age": 26,
    "department": "Cloud",
    "skills": ["aws", "docker"],
    "address": { "city": "Kochi", "state": "Kerala" },
    "projects": [
      { "name": "USTCloudPortal", "status": "planned" }
    ],
    "experience_years": 3
```

```
},
{
  "emp_id": 103,
  "name": "Sahana R",
  "age": 22,
  "department": "Testing",
  "skills": ["selenium"],
  "address": { "city": "Chennai", "state": "TN" },
  "projects": [],
  "experience_years": 1
},
{
  "emp_id": 104,
  "name": "Vishnu Prakash",
  "age": 29,
  "department": "Cybersecurity",
  "skills": ["networking", "siem"],
  "address": { "city": "Trivandrum", "state": "Kerala" },
  "projects": [
    { "name": "SecureBank", "status": "in-progress" }
  ],
  "experience_years": 5
},
{
  "emp_id": 105,
  "name": "Maya Kumar",
  "age": 25,
  "department": "AI",
  "skills": ["python", "deep-learning"],
  "address": { "city": "Bangalore", "state": "KA" },
  "projects": [
    { "name": "VisionAI", "status": "completed" }
  ],
  "experience_years": 2
},
{
```

```
  "emp_id": 106,
  "name": "Arjun S",
  "age": 28,
  "department": "Cloud",
  "skills": ["azure"],
  "address": { "city": "Kochi", "state": "Kerala" },
  "projects": [
    { "name": "DevOpsBoost", "status": "in-progress" }
  ],
  "experience_years": 4
},
{
  "emp_id": 107,
  "name": "Neha Mohan",
  "age": 24,
  "department": "Data",
  "skills": ["sql", "tableau"],
  "address": { "city": "Chennai", "state": "TN" },
  "projects": [
    { "name": "USTAnalyticsHub", "status": "planned" }
  ],
  "experience_years": 2
},
{
  "emp_id": 108,
  "name": "Suresh B",
  "age": 27,
  "department": "Data",
  "address": { "city": "Kochi", "state": "Kerala" },
  "projects": [],
  "experience_years": 3
},
{
  "emp_id": 109,
  "name": "Lavanya N",
  "age": 21,
```

```
    "department": "Testing",
    "skills": [],
    "address": { "city": "Trivandrum", "state": "Kerala" },
    "projects": [
      { "name": "QualityX", "status": "in-progress" }
    ],
    "experience_years": 0
  },
  {
    "emp_id": 110,
    "name": "Rakesh Pillai",
    "age": 30,
    "department": "Cybersecurity",
    "skills": ["networking", "firewall", "linux"],
    "projects": [
      { "name": "SecuritySuite", "status": "deployed" }
    ],
    "experience_years": 7
  }
]
```